

Netlists

OTA_5T:

AC:

```
* 5T-OTA - Parallel AC Testbench (DM + CM)
Vdd Vdd 0 3

* =====
* 1. BRANCH 1: DIFFERENTIAL MODE (DM)
* =====
IB_dm Vdd vb_dm {Ib}

* Inputs DM
vicm_dm vcm_dm 0 {Vicm}
vinp_dm vip_dm vcm_dm SINE(0 10u 30) AC 0.5
vinn_dm vin_dm vcm_dm SINE(0 -10u 30) AC -0.5

* Circuit DM
M1_dm vm_dm vip_dm vT_dm 0 nmos_3p3 l={Ld} w={Wd}
M2_dm vout_dm vin_dm vT_dm 0 nmos_3p3 l={Ld} w={Wd}
M3_dm vm_dm vm_dm Vdd Vdd pmos_3p3 l={Lm} w={Wm}
M4_dm vout_dm vm_dm Vdd Vdd pmos_3p3 l={Lm} w={Wm}
M5_dm vT_dm vb_dm 0 0 nmos_3p3 l={LSS} w={Wss}
M6_dm vb_dm vb_dm 0 0 nmos_3p3 l={LSS} w={WB}
CL_dm vout_dm 0 {CL}

* =====
* 2. BRANCH 2: COMMON MODE (CM)
* =====
IB_cm Vdd vb_cm {Ib}

* Inputs CM
vicm_ac vicm_ac 0 {Vicm} AC 1

* Circuit CM
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M1_cm vm_cm vicm_ac vT_cm 0 nmos_3p3 l={Ld} w={Wd}
M2_cm vout_cm vicm_ac vT_cm 0 nmos_3p3 l={Ld} w={Wd}
M3_cm vm_cm vm_cm Vdd Vdd pmos_3p3 l={Lm} w={Wm}
M4_cm vout_cm vm_cm Vdd Vdd pmos_3p3 l={Lm} w={Wm}
M5_cm vT_cm vb_cm 0 0 nmos_3p3 l={LSS} w={Wss}
M6_cm vb_cm vb_cm 0 0 nmos_3p3 l={LSS} w={WB}
CL_cm vout_cm 0 {CL}

* =====
* MODELS & PARAMS
* =====

.model NMOS NMOS
.model PMOS PMOS
.lib C:\Users\Mohamed\AppData\Local\LTspice\lib\cmp\standard.mos
.inc cmos_tt.txt

.param Ld=1e-06
.param Wd=2e-05
.param Lm=1e-06
.param Wm=1e-05
.param LSS=1e-06
.param WSS=2e-05
.param WB=1e-05
.param Ib=1e-05
.param vicm=2.5
.param CL=1e-13

* Analysis
.ac dec 100 1k 1e9

* Measurements
.MEAS AC AVD max mag(V(vout_dm))
.MEAS AC BW WHEN MAG(V(vout_dm)) = AVD/sqrt(2)
.MEAS AC UGF WHEN MAG(V(vout_dm)) = 1

.MEAS AC AVCM FIND mag(V(vout_cm)) AT 1k
.MEAS AC CMRR param AVD/AVCM

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.end
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DC:

```
* OTA_5T
Vdd Vdd 0 3

* Inputs
Vicm vcm 0 {Vicm}
vinp vip vcm SINE(0 10u 30) AC 0.5
vinn vin vcm SINE(0 -10u 30) AC -0.5

* Current Reference
IB Vdd vb {Ib}

* Cap
CL vout 0 {CL}

* Diff pair
M1 vm vip vT 0 nmos_3p3 l={Ld} w={Wd}
M2 vout vin vT 0 nmos_3p3 l={Ld} w={Wd}

*Mirror
M3 vm vm Vdd Vdd pmos_3p3 l={Lm} w={Wm}
M4 vout vm Vdd Vdd pmos_3p3 l={Lm} w={Wm}

* Tail and bias
M5 vT vb 0 0 nmos_3p3 l={LSS} w={Wss}
M6 vb vb 0 0 nmos_3p3 l={LSS} w={WB}
.lib C:\Users\Mohamed\AppData\Local\LTspice\lib\cmp\standard.mos
.op
```

```

* libray
.inc cmos_tt.txt

.param Ld=1e-06
.param Wd=2e-05
.param Lm=1e-06
.param Wm=1e-05
.param LSS=1e-06
.param WSS=2e-05
.param WB=1e-05
.param Ib=1e-05
.param vicm=2.5
.param CL=1e-13

*.meas OP Idd PARAM abs(I(Vdd))
.meas OP Power PARAM abs(V(Vdd)*I(Vdd))
.meas OP VOUT_CM PARAM V(vout)
.meas OP SLEW_RATE PARAM abs(Id(M5))/CL
.end

```

Buffer:

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* OTA_5T
Vdd Vdd 0 3

* Inputs
vinp vip 0 {vicm}

* Current Reference
IB Vdd vb {Ib}

* Cap
CL vout 0 {CL}

```

```

* Diff pair
M1 vm vip vT 0 nmos_3p3 l={Ld} w={Wd}
M2 vout vout vT 0 nmos_3p3 l={Ld} w={Wd}

*Mirror
M3 vm vm Vdd Vdd pmos_3p3 l={Lm} w={Wm}
M4 vout vm Vdd Vdd pmos_3p3 l={Lm} w={Wm}

* Tail and bias
M5 vT vb 0 0 nmos_3p3 l={LSS} w={Wss}
M6 vb vb 0 0 nmos_3p3 l={LSS} w={WB}
.lib C:\Users\Mohamed\AppData\Local\LTspice\lib\cmp\standard.mos
.op

* libray
.inc cmos_tt.txt

.param Ld=1e-06
.param Wd=2e-05
.param Lm=1e-06
.param Wm=1e-05
.param LSS=1e-06
.param WSS=2e-05
.param WB=1e-05
.param Ib=1e-05
.param vicm=2.5
.param CL=1e-13

*.meas OP Idd PARAM abs(I(Vdd))
.meas OP VOUT_ERR PARAM V(vout)-v(vip)

.end

```

Miller OTA :

AC

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*=====
* PMOS-Input Miller OTA - Parallel AC Testbench (DM + CM)
*=====

Vdd Vdd 0 3

* =====
* 1. BRANCH 1: DIFFERENTIAL MODE (DM)
* =====
* Input Stimulus (Differential AC Superimposed on Vicm)
vicm_dm vcm_dm 0 {vicm}
vinp_dm vip_dm vcm_dm AC 0.5
vinn_dm vin_dm vcm_dm AC -0.5

* Bias Circuit (PMOS Current Mirror Base)
IB_dm vb_dm 0 {Ib}
M8_dm vb_dm vb_dm Vdd Vdd pmos_3p3 l={Lss} w={WB}

* First Stage (PMOS Differential Pair + NMOS Active Load)
M5_dm tail_dm vb_dm Vdd Vdd pmos_3p3 l={Lss} w={Wss}
M1_dm vx_dm vin_dm tail_dm Vdd pmos_3p3 l={Ld} w={Wd}
M2_dm v1_dm vip_dm tail_dm Vdd pmos_3p3 l={Ld} w={Wd}
M3_dm vx_dm vx_dm 0 0 nmos_3p3 l={Lm} w={Wm}
M4_dm v1_dm vx_dm 0 0 nmos_3p3 l={Lm} w={Wm}

* Second Stage (NMOS Common Source + PMOS Active Load)
M7_dm vout_dm vb_dm Vdd Vdd pmos_3p3 l={Lss} w={Wss*beta}
M6_dm vout_dm v1_dm 0 0 nmos_3p3 l={Lm} w={Wm*2*beta}

* Compensation Network
Rz_dm v1_dm v_comp_dm {Rz}
Cc_dm v_comp_dm vout_dm {Cc}

* Load Capacitance
CL_dm vout_dm 0 {CL}

* =====
* 2. BRANCH 2: COMMON MODE (CM)
```

```

* =====
* Input Stimulus (Common AC Superimposed on Vicm)
vicm_ac vicm_ac 0 {vicm} AC 1

* Bias Circuit (PMOS Current Mirror Base)
IB_cm vb_cm 0 {Ib}
M8_cm vb_cm vb_cm Vdd Vdd pmos_3p3 l={Lss} w={WB}

* First Stage (PMOS Differential Pair + NMOS Active Load)
M5_cm tail_cm vb_cm Vdd Vdd pmos_3p3 l={Lss} w={Wss}
M1_cm vx_cm vicm_ac tail_cm Vdd pmos_3p3 l={Ld} w={Wd}
M2_cm v1_cm vicm_ac tail_cm Vdd pmos_3p3 l={Ld} w={Wd}
M3_cm vx_cm vx_cm 0 0 nmos_3p3 l={Lm} w={Wm}
M4_cm v1_cm vx_cm 0 0 nmos_3p3 l={Lm} w={Wm}

* Second Stage (NMOS Common Source + PMOS Active Load)
M7_cm vout_cm vb_cm Vdd Vdd pmos_3p3 l={Lss} w={Wss*beta}
M6_cm vout_cm v1_cm 0 0 nmos_3p3 l={Lm} w={Wm*2*beta}

* Compensation Network
Rz_cm v1_cm v_comp_cm {Rz}
Cc_cm v_comp_cm vout_cm {Cc}

* Load Capacitance
CL_cm vout_cm 0 {CL}

* =====
* MODELS & PARAMS
* =====
.model NMOS NMOS
.model PMOS PMOS
.inc cmos_tt.txt

* Device Parameters (Formatted one per line for the Python parser)
.param Ld=1e-06
.param Wd=2e-05

.param Lm=1e-06
.param Wm=1e-05

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.param LSS=1e-06
.param WSS=2e-05

.param WB=1e-05

.param beta=1

.param Ib=1e-05

.param Cc=1e-12
.param Rz=5000

.param CL=1e-13

.param vicm=2

* =====
* ANALYSIS & MEASUREMENTS
* =====

.ac dec 100 1 10G

* Differential Mode Measurements
.MEAS AC AVD MAX MAG(V(vout_dm))
.MEAS AC BW WHEN MAG(V(vout_dm)) = AVD/sqrt(2)
.MEAS AC UGF WHEN MAG(V(vout_dm)) = 1

* Common Mode & CMRR Measurements
.MEAS AC AVCM FIND MAG(V(vout_cm)) AT 1
.MEAS AC CMRR param AVD/AVCM

.end

```



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*=====
* PMOS-Input Miller OTA - AC Analysis Testbench
*=====

Vdd Vdd 0 3

* Input Stimulus (Differential AC Superimposed on Vicm)
Vicm vicm 0 {vicm}
Vinp vip vicm AC 0.5
Vinn vin vicm AC -0.5

* Bias Circuit (PMOS Current Mirror Base)
* Note: Current flows from VDD through M8 to Ground via IB
IB vb 0 {Ib}
M8 vb vb Vdd Vdd pmos_3p3 l={Lss} w={WB}

* First Stage (PMOS Differential Pair + NMOS Active Load)
M5 tail vb Vdd Vdd pmos_3p3 l={Lss} w={Wss}
M1 vx vin tail Vdd pmos_3p3 l={Ld} w={Wd}
M2 v1 vip tail Vdd pmos_3p3 l={Ld} w={Wd}
M3 vx vx 0 0 nmos_3p3 l={Lm} w={Wm}
M4 v1 vx 0 0 nmos_3p3 l={Lm} w={Wm}

* Second Stage (NMOS Common Source + PMOS Active Load)
* Dimensions scale automatically based on the 'beta' parameter
M7 vout vb Vdd Vdd pmos_3p3 l={Lss} w={Wss*beta}
M6 vout v1 0 0 nmos_3p3 l={Lm} w={Wm*2*beta}

* Compensation Network
Rz v1 v_comp {Rz}
Cc v_comp vout {Cc}

* Load Capacitance
CL vout 0 {CL}

* Models & Libs
.model NMOS NMOS
.model PMOS PMOS
.inc cmos_tt.txt

```

* Analysis Commands

.op

* Device Parameters (Formatted one per line for the Python parser)

.param Ld=1e-06

.param Wd=2e-05

.param Lm=1e-06

.param Wm=1e-05

.param LSS=1e-06

.param WSS=2e-05

.param WB=1e-05

.param beta=1

.param Ib=1e-05

.param Cc=1e-12

.param Rz=5000

.param CL=1e-13

.param vicm=1.5

* Automated AC Measurements

.meas OP Power PARAM abs(V(Vdd)*I(Vdd))

.meas OP VOUT_CM PARAM V(vout)

.meas OP SLEW_RATE PARAM abs(Id(M5))/Cc

.end

Buffer

*=====

* PMOS-Input Miller OTA - AC Analysis Testbench

*=====

Vdd Vdd 0 3

* Input Stimulus (Differential AC Superimposed on Vicm)

Vinp vip 0 {vicm}

* Bias Circuit (PMOS Current Mirror Base)

* Note: Current flows from VDD through M8 to Ground via IB

IB vb 0 {Ib}

M8 vb vb Vdd Vdd pmos_3p3 l={Lss} w={WB}

* First Stage (PMOS Differential Pair + NMOS Active Load)

M5 tail vb Vdd Vdd pmos_3p3 l={Lss} w={Wss}

M1 vx vout tail Vdd pmos_3p3 l={Ld} w={Wd}

M2 v1 vip tail Vdd pmos_3p3 l={Ld} w={Wd}

M3 vx vx 0 0 nmos_3p3 l={Lm} w={Wm}

M4 v1 vx 0 0 nmos_3p3 l={Lm} w={Wm}

* Second Stage (NMOS Common Source + PMOS Active Load)

* Dimensions scale automatically based on the 'beta' parameter

M7 vout vb Vdd Vdd pmos_3p3 l={Lss} w={Wss*beta}

M6 vout v1 0 0 nmos_3p3 l={Lm} w={Wm*2*beta}

* Compensation Network

Rz v1 v_comp {Rz}

Cc v_comp vout {Cc}

* Load Capacitance

CL vout 0 {CL}

* Models & Libs

.model NMOS NMOS

.model PMOS PMOS

.inc cmos_tt.txt

* Analysis Commands

.op

* Device Parameters (Formatted one per line for the Python parser)

.param Ld=1e-06

.param Wd=2e-05

.param Lm=1e-06

.param Wm=1e-05

.param LSS=1e-06

.param WSS=2e-05

.param WB=1e-05

```
.param beta=1
.param Ib=1e-05
.param Cc=1e-12
.param Rz=5000
.param CL=1e-13
.param vicm=2.5

* Automated AC Measurements
.meas OP VOUT_ERR PARAM V(vout)-V(vip)

.end
```